powder you started with. Now, you can be an instrumentalist—you can deny that electron orbitals are real things that are involved in this reaction—but then how can you explain the fabulous match between the theoretical prediction and the experimental outcome? Why does quantum chemistry work so well to explain the thermite reaction if atoms and electron orbitals aren't real? "If [instrumentalism] is correct we must believe in a *cosmic coincidence*," said J. J. C. Smart. "Is it not odd that the phenomena of the world should be such as to make a purely instrumental theory true? On the other hand, if we interpret a theory in a realist way, then we have no need for such a cosmic coincidence… a lot of surprising facts no longer seem surprising." Smart went on, clearly impatient with instrumentalism:

> Suppose that a detective finds a lot of footprints, bloodstains, and so on. If the criminal were a theoretical fiction for relating footprints and bloodstains hitherto found to one another, then it would seem too good to be true that it should actually issue in true predictions of further footprints and bloodstains and even of missing five-pound notes. But if there really were a criminal, then these predictions would no longer be surprising.

Hilary Putnam put it more succinctly. "Realism," he claimed, "is the only philosophy that doesn't make the success of science a miracle."

Smart didn't just see positivism as problematic philosophically—inspired by Feyerabend, he saw it as a practical problem as well. "A positivist attitude has frequently been inimical to progress," wrote Smart in 1963. "Positivism would once have supported the Ptolemaic [Earth-centered] theory against the Copernican one, by showing that at the time it was the better [at making predictions] of the two. It supported phenomenological thermodynamics and resisted the [atomic] theory of gases. And today it opposes, *a priori*, any attempts to construct alternatives to the prevailing Copenhagen interpretation of quantum mechanics." This, to Smart, and to Putnam, Feyerabend, and other leading philosophers of the day, was a serious problem—because with the collapse of positivism, the Copenhagen interpretation was indefensible. How could our everyday world of things